



Switching & Routing

Xiaofeng Zhong



1



Routing



2



Forwarding Decisions

⊕ When packet arrives..

- ❑ Must decide which outgoing port to use
- ❑ In single transmission time
- ❑ Forwarding decisions must be *simple*

⊕ Routing state dictates where to forward packets

- ❑ Assume decisions are *deterministic*

⊕ *Global routing state* means collection of routing state in each of the routers

- ❑ Will focus on where this routing state comes from
- ❑ But first, a few preliminaries....

3



Outline (Routing)

Techniques

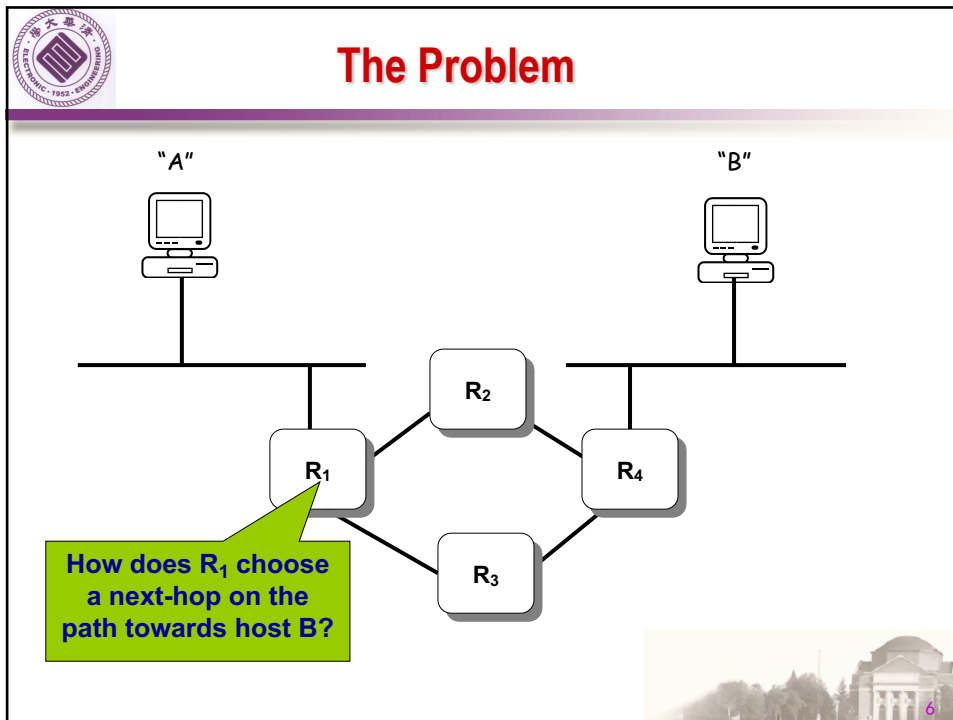
- ❖ Naïve: Flooding
- ❖ Distance vector: Distributed Bellman Ford Algorithm
- ❖ Link state: Dijkstra's Shortest Path First-based Algorithm

Routing in the Internet

- ❖ Hierarchy and Autonomous Systems
- ❖ Interior Routing Protocols: RIP, OSPF
- ❖ Exterior Routing Protocol: BGP

Routing is a very complex subject, and has many aspects.
Here, we will concentrate on the basics.

5



6

 **Introduction (Routing)**

⊕ Routing

- Routing refers to the network-wide process that determines the end-to-end paths that packets take from source to destination

⊕ Routing Algorithm

- Determine the least-cost paths from senders to receivers through the network of routers
- Function 1: determine the route taken by packets as they flow from a sender to a receiver
- Function 2: transmit the information related with routing, e.g., the state of links and nodes



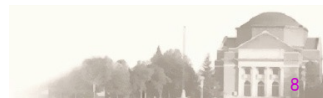
7



Routing Metrics

⊕ Metrics

- ❑ Delay to send an average size packet (Make high speed links attractive, but closeness counts)
- ❑ Bandwidth
- ❑ Link utilization
- ❑ Stability: Is a link (or path) up or down?

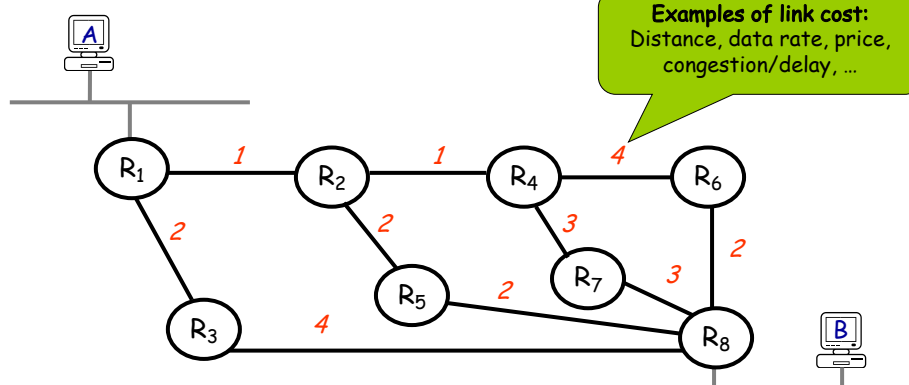


8



Example network

Objective: Determine the route from A to B that minimizes the path cost.



9

Example network

In this simple case, solution is clear from inspection

10

So what about this network...!?

The public Internet in 1999

Lucent Technologies
©1999 Lucent Technologies

Learn more at
<http://www.lumeta.com>

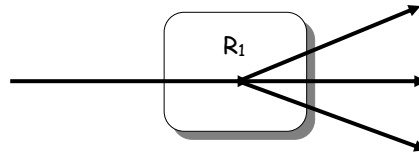
Today: about 1/3 of Internet routes are asymmetric

11



Technique 1: Naïve Approach

Flood! -- Routers forward packets to all ports except the ingress port.



Advantages:

- ❖ Simple.
- ❖ Every destination in the network is reachable.

Disadvantages:

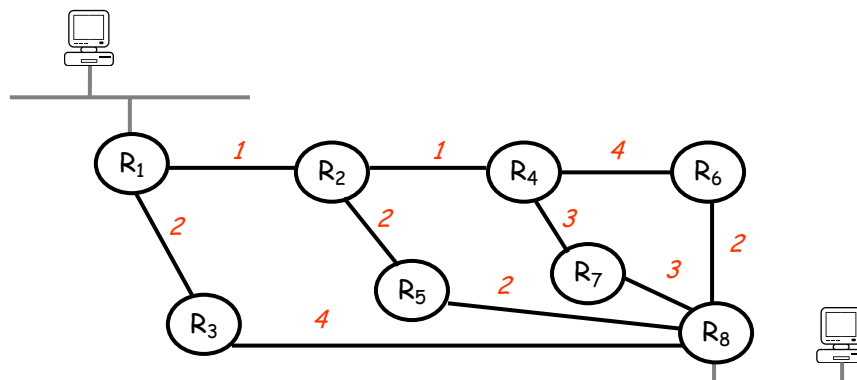
- ❖ Some routers receive a packet multiple times.
- ❖ Packets can go round in loops forever.
- ❖ Inefficient.

12



Spanning Trees

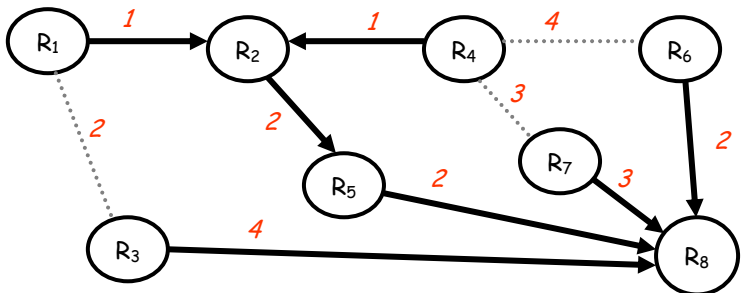
Objective: Find the lowest cost route from each of $(R_1, \dots, R_7)$ to R_8 .



13



A Spanning Tree



- ❖ The solution is a **spanning tree** with R_8 as the root of the tree.
- ❖ **Tree:** There are no loops.
- ❖ **Spanning:** All nodes included.
- ❖ We'll see two algorithms that build spanning trees automatically:
 - ❖ The distributed Bellman-Ford algorithm
 - ❖ Dijkstra's shortest path first algorithm



14

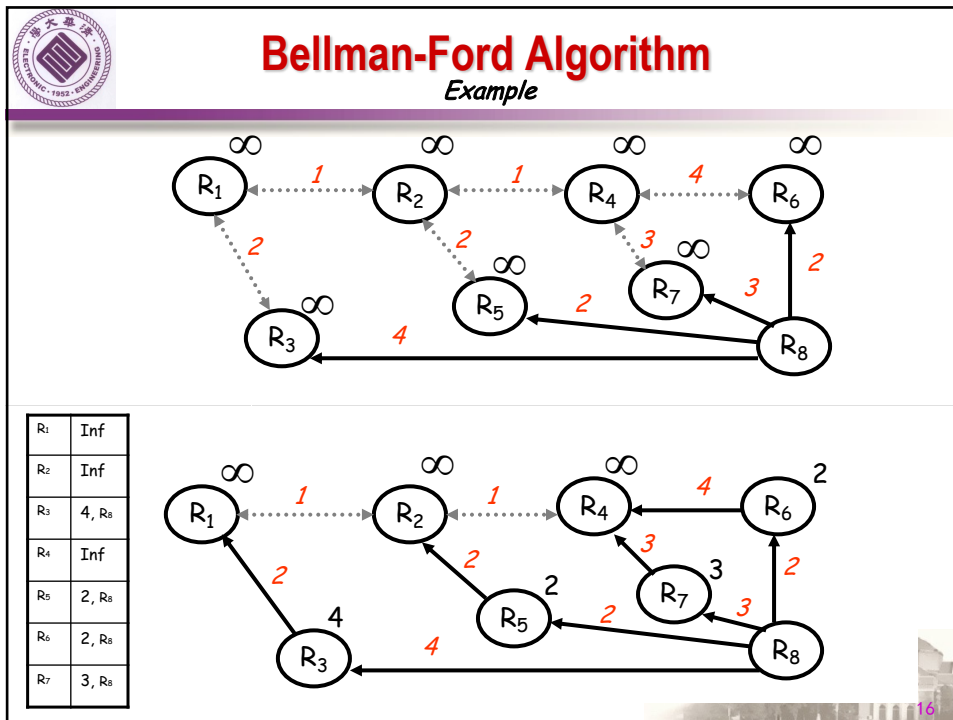


Technique 2: Distance Vector The Distributed Bellman-Ford Algorithm

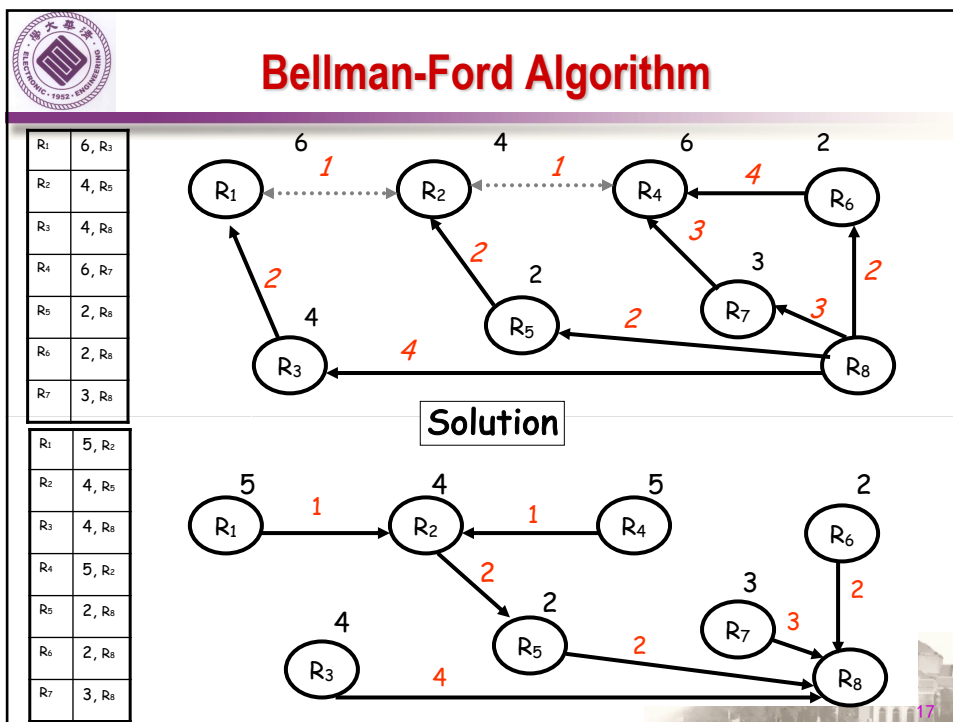
1. Let $\underline{X}_n = (C_1, C_2, \dots, C_7)$ where: $C_i = \text{cost from } R_i \text{ to } R_8$.
2. Set $\underline{X}_0 = (\infty, \infty, \infty, \dots, \infty)$.
3. Every T seconds, router i sends C_i , for all i , to its neighbors. This is the "Distance vector".
4. If router i is told of a lower cost path to R_8 , it updates C_i . Hence, $\underline{X}_{n+1} = f(\underline{X}_n)$ where $f(\cdot)$ determines the next step improvement.
5. If $\underline{X}_{n+1} \neq \underline{X}_n$ then goto step (3).
6. Stop.



15



16

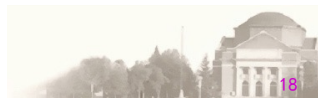


17



Technique 3: Link State Dijkstra's Shortest Path Algorithm

- ⊕ Whenever the state of an incident link changes,
 - ⊕ Routers send out update messages
 - ❑ Called "Link State Updates"
- ⊕ Based on all link state updates received,
 - ⊕ each router calculates lowest cost path to all others, starting from itself.
 - ❑ Use Dijkstra's single-source shortest path algorithm
 - ❑ Assume all updates are consistent
- ⊕ At each step of the algorithm, router adds the next shortest (i.e. lowest-cost) path to the tree.
- ⊕ Finds spanning tree rooted at the router.



18



Reliable Flooding of LSP

- ⊕ The Link State Packet (LSP):
 - ❑ The ID of the router that created the LSP
 - ❑ List of directly connected neighbors, and cost
 - ❑ Sequence number
 - ❑ TTL
- ⊕ Reliable Flooding
 - ❑ Check the sequence number
 - newer, resend LSP over all links other than incident link,
 - Otherwise, drop it.
- ⊕ Link State Detection:
 - ❑ Link layer failure
 - ❑ Loss of "hello" packets



19



Link-State Routing Is Conceptually Simple

- ⊕ Each router keeps track of its incident links
- ⊕ Each router broadcasts the link state
 - To give every router a complete view of the graph
- ⊕ Each router runs Dijkstra's algorithm
 - Compute shortest paths, then construct forwarding table
- ⊕ Example protocols
 - Open Shortest Path First (OSPF)
 - Intermediate System – Intermediate System (IS-IS)
- ⊕ Challenges: scaling, transient disruptions



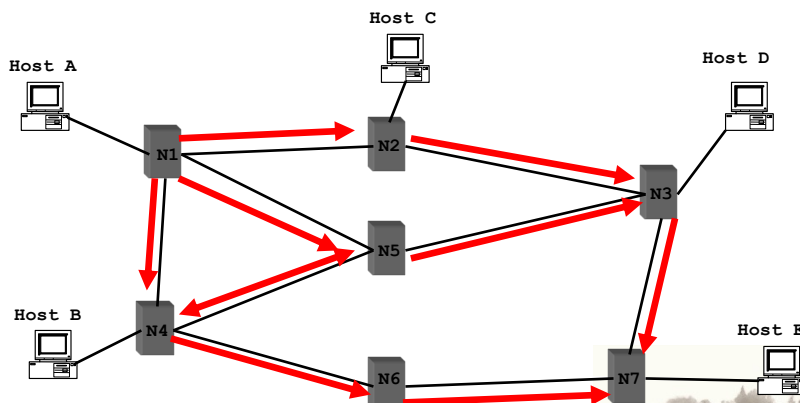
20

20



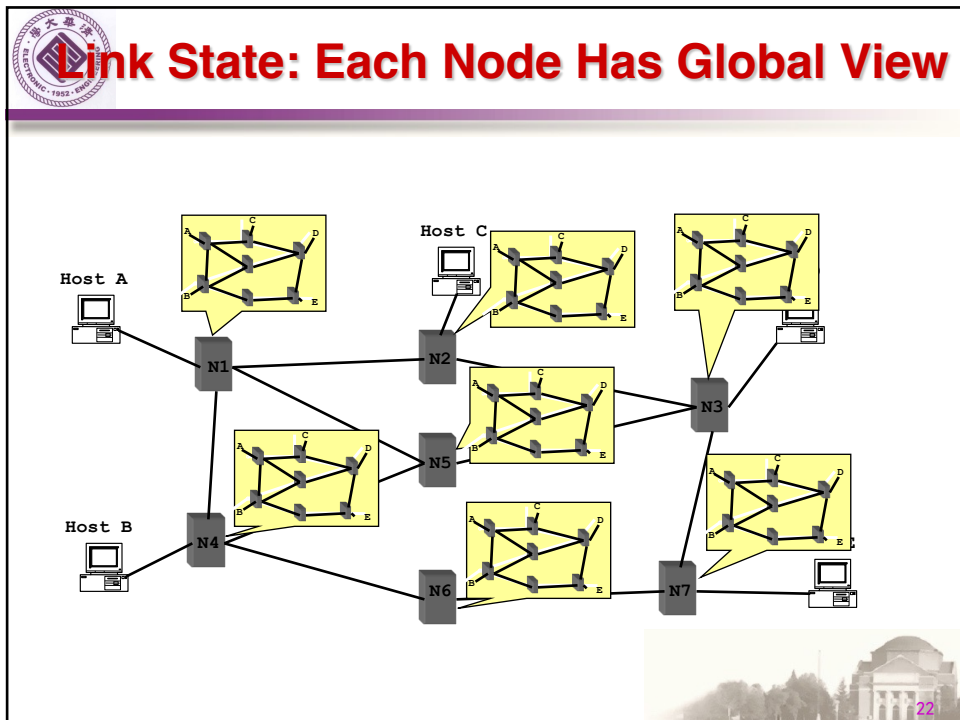
Link State Routing

- ⊕ Each node floods its local information
- ⊕ Each node then knows **entire** network topology



21

21



22

How to Compute Routes

- ⊕ Each node should have **same** global view
- ⊕ They each compute their own routing tables
- ⊕ Using *exactly* the same algorithm
- ⊕ Can use *any* algorithm that avoids loops
- ⊕ Computing shortest paths is one such algorithm
 - ▣ Associate a “cost” with each link
 - ▣ Don’t worry what it stands for....

23



Dijkstra's Shortest Path Algorithm

⊕ INPUT:

- Network topology (graph), with link costs

⊕ OUTPUT:

- Least cost paths from one node to all other nodes
- Produces “tree” of routes
 - Different from what we talked about before
 - Previous tree was rooted at destination
 - This is rooted at source
 - But shortest paths are reversible!



24

24



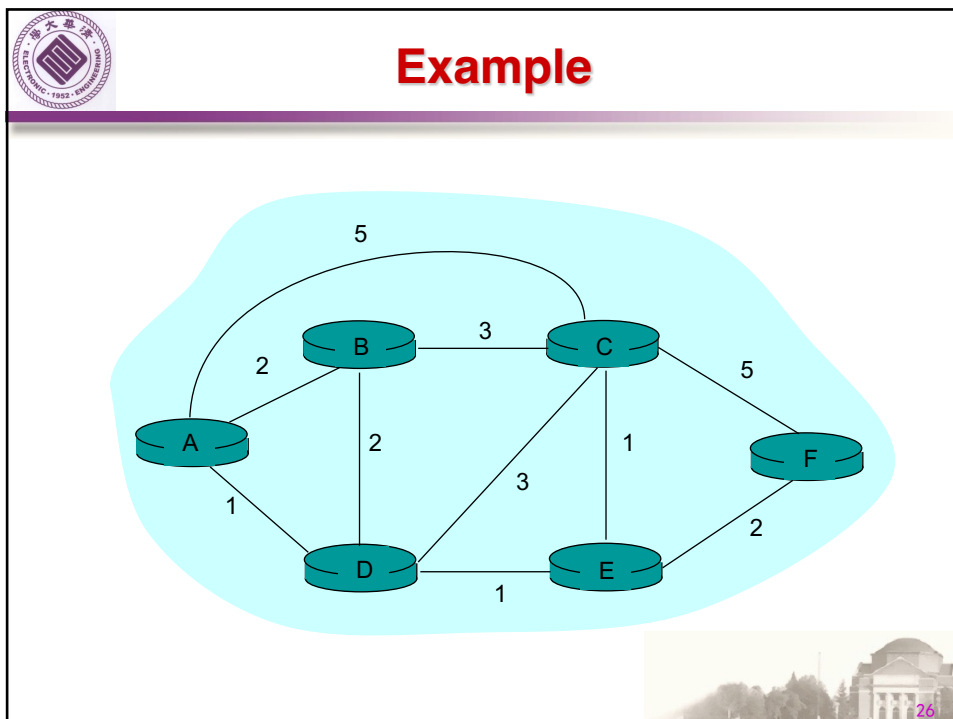
“Least Cost” Routes

- ⊕ No sensible cost metric will be minimized by traversing a loop
- ⊕ “Least cost” routes an easy way to avoid loops
- ⊕ Least cost routes are also “destination-based”
 - i.e., do not depend on the source
 - Why is this?
- ⊕ Therefore, least-cost paths form a spanning tree

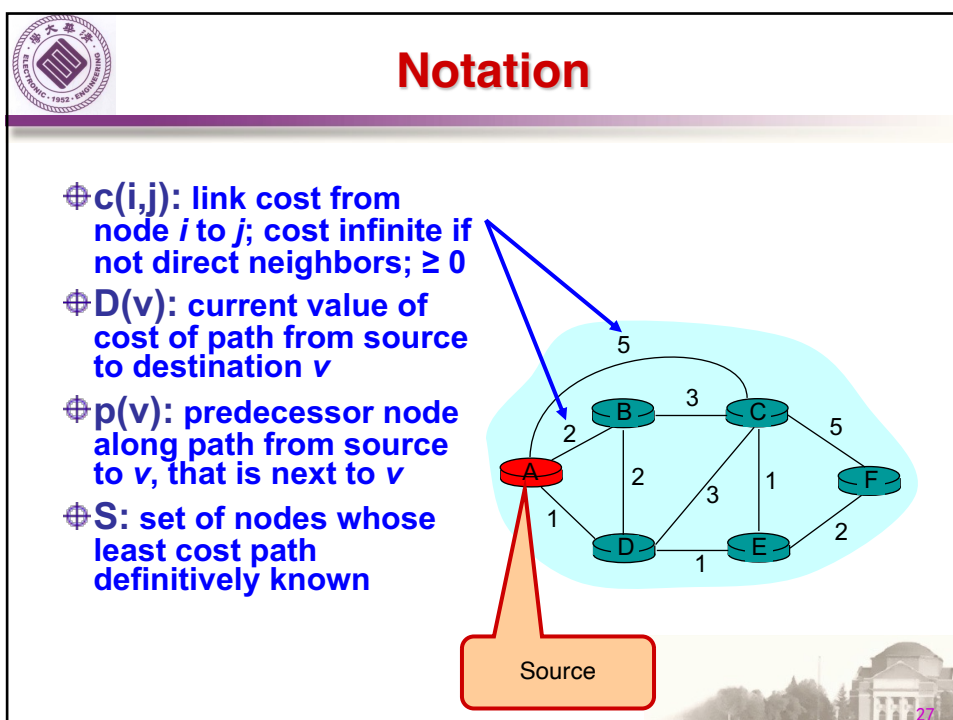


25

25



26



- ⊕ $c(i,j)$: link cost from node i to j ; cost infinite if not direct neighbors; ≥ 0
- ⊕ $D(v)$: current value of cost of path from source to destination v
- ⊕ $p(v)$: predecessor node along path from source to v , that is next to v
- ⊕ S : set of nodes whose least cost path definitively known

27



Dijkstra's Algorithm

1 Initialization:

```

2  $S = \{A\};$ 
3 for all nodes  $v$ 
4   if  $v$  adjacent to  $A$ 
5     then  $D(v) = c(A,v);$ 
6     else  $D(v) = \infty;$ 
7

```

- $c(i,j)$: link cost from node i to j
- $D(v)$: current cost source $\rightarrow v$
- $p(v)$: predecessor node along path from source to v , that is next to v
- S : set of nodes whose least cost path definitively known

8 Loop

```

9 find  $w$  not in  $S$  such that  $D(w)$  is a minimum;
10 add  $w$  to  $S$ ;
11 update  $D(v)$  for all  $v$  adjacent to  $w$  and not in  $S$ :
12   if  $D(w) + c(w,v) < D(v)$  then
13     //  $w$  gives us a shorter path to  $v$  than we've found so far
14      $D(v) = D(w) + c(w,v); p(v) = w;$ 
15 until all nodes in  $S$ ;

```

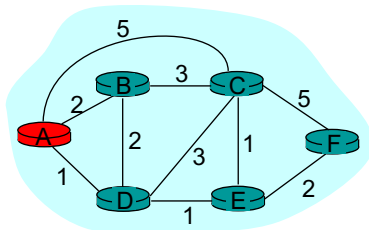


28



Example: Dijkstra's Algorithm

Step	start S	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	$D(E), p(E)$	$D(F), p(F)$
0	A	2, A	5, A	1, A	∞	∞
1						
2						
3						
4						
5						



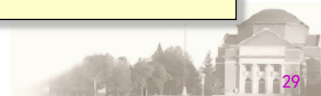
1 Initialization:

```

2  $S = \{A\};$ 
3 for all nodes  $v$ 
4   if  $v$  adjacent to  $A$ 
5     then  $D(v) = c(A,v);$ 
6     else  $D(v) = \infty;$ 
...

```

$D(v)$: current value of cost of path from source to destination v

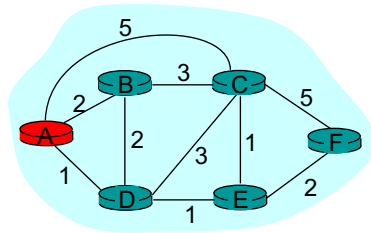


29



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1						
2						
3						
4						
5						



```

...
8  Loop
9  find w not in S s.t. D(w) is a minimum;
10 add w to S;
11 update D(v) for all v adjacent
    to w and not in S:
12 If D(w) + c(w,v) < D(v) then
13   D(v) = D(w) + c(w,v); p(v) = w;
14 until all nodes in S;

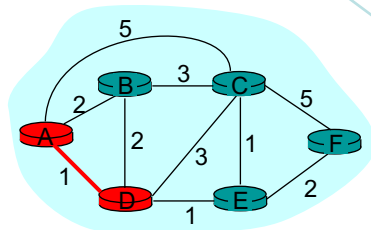
```

30



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD					
2						
3						
4						
5						



```

...
8  Loop
9  find w not in S s.t. D(w) is a minimum;
10 add w to S;
11 update D(v) for all v adjacent
    to w and not in S:
12 If D(w) + c(w,v) < D(v) then
13   D(v) = D(w) + c(w,v); p(v) = w;
14 until all nodes in S;

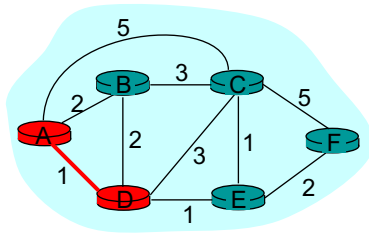
```

31



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2						
3						
4						
5						



```

...
8 Loop
9   find w not in S s.t.  $D(w)$  is a minimum;
10  add w to S;
11  update  $D(v)$  for all v adjacent
    to w and not in S:
12  If  $D(w) + c(w,v) < D(v)$  then
13     $D(v) = D(w) + c(w,v)$ ;  $p(v) = w$ ;
14  until all nodes in S;

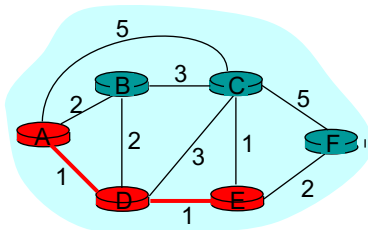
```

32



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3						
4						
5						



```

...
8 Loop
9   find w not in S s.t.  $D(w)$  is a minimum;
10  add w to S;
11  update  $D(v)$  for all v adjacent
    to w and not in S:
12  If  $D(w) + c(w,v) < D(v)$  then
13     $D(v) = D(w) + c(w,v)$ ;  $p(v) = w$ ;
14  until all nodes in S;

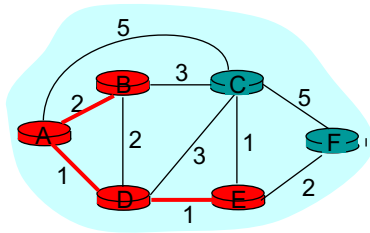
```

33



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4						
5						



```

...
8 Loop
9   find w not in S s.t. D(w) is a minimum;
10  add w to S;
11  update D(v) for all v adjacent
    to w and not in S:
12  If D(w) + c(w,v) < D(v) then
13    D(v) = D(w) + c(w,v); p(v) = w;
14  until all nodes in S;

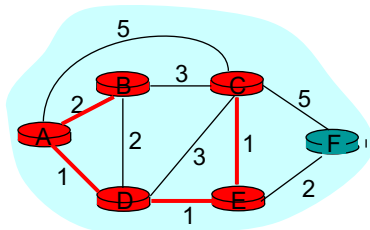
```

34



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5						



```

...
8 Loop
9   find w not in S s.t. D(w) is a minimum;
10  add w to S;
11  update D(v) for all v adjacent
    to w and not in S:
12  If D(w) + c(w,v) < D(v) then
13    D(v) = D(w) + c(w,v); p(v) = w;
14  until all nodes in S;

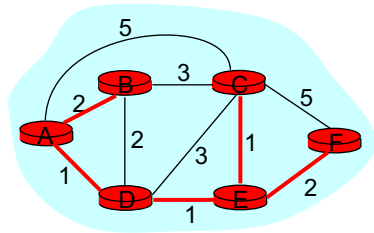
```

35



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5	ADEBCF					



```

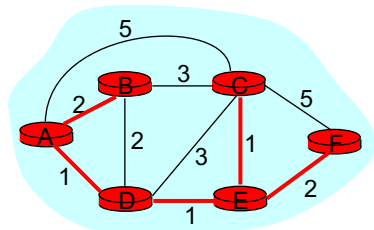
...
8 Loop
9   find w not in S s.t. D(w) is a minimum;
10  add w to S;
11  update D(v) for all v adjacent
    to w and not in S:
12  If D(w) + c(w,v) < D(v) then
13    D(v) = D(w) + c(w,v); p(v) = w;
14  until all nodes in S;
  
```

36



Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5	ADEBCF					



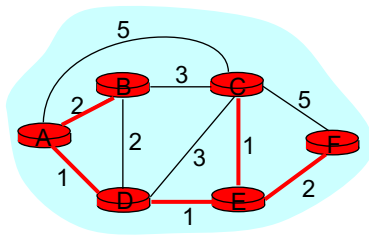
To determine path $A \rightarrow C$ (say),
work backward from C via $p(v)$

37



The Forwarding Table

- Running Dijkstra at node A gives the shortest path from A to all destinations
- We then construct the *forwarding table*



Destination	Link
B	(A,B)
C	(A,D)
D	(A,D)
E	(A,D)
F	(A,D)

38



Distance-Vector vs. Link-State

D-V

- asynchronous, and distributed algorithm.
- Maintain routing to all destinations, but only next hop ID and cost, not complete path
- Periodically update route table (whole table)
- Typical algorithm: Bellman-Ford algorithm

L-S

- Keep the whole network topology
- Calculate shortest routing path, while need
- Event trigger, update link stage change
- Typical algorithm: Dijkstra's algorithm

39



Routing

□ Introduction

□ Routing algorithm

- Distance-vector routing algorithm
- Link-state routing algorithm
- Multicast routing

□ Routing protocol

- Intra-AS routing
- Inter-AS routing



40

40



Routing in the Internet

The Internet uses hierarchical routing

⊕ The Internet is split into Autonomous Systems (AS's)

- ❖ Examples of AS's: Stanford (32), HP (71), MCI Worldcom (17373)
- ❖ Try: `whois -h whois.arin.net "MCI Worldcom"`

⊕ Within an AS, the administrator chooses an Interior Gateway Protocol (IGP)

- ❖ Examples of IGPs: RIP (rfc 1058, **D-V**), OSPF (rfc 1247, **L-S**).

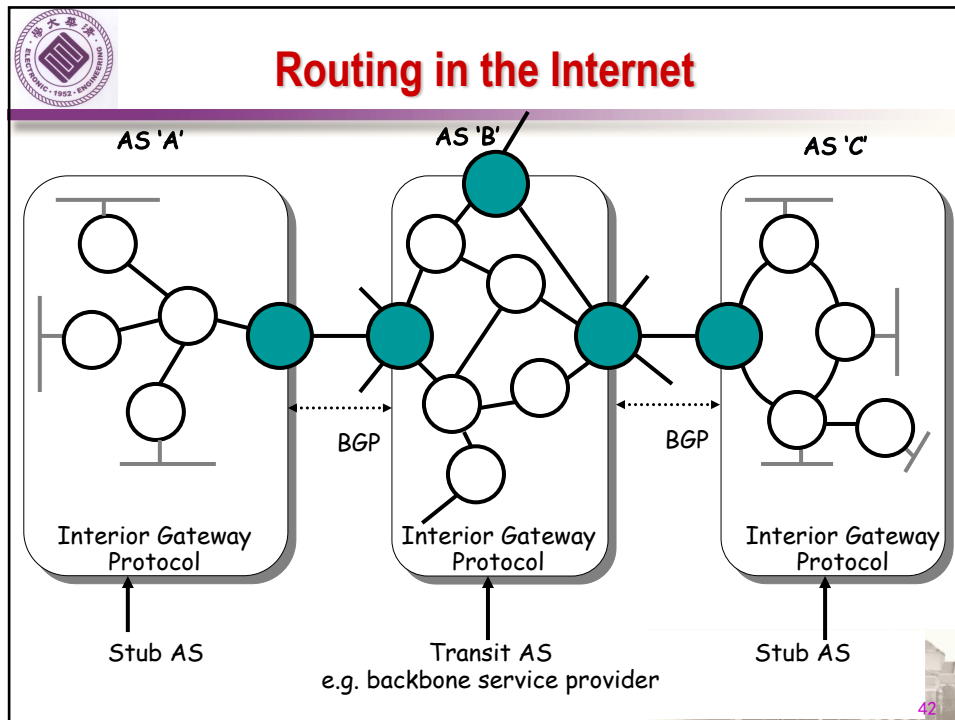
⊕ Between AS's, the Internet uses an Exterior Gateway Protocol

- ❖ AS's today use the Border Gateway Protocol, BGP-4 (rfc 1771, **D-V&L-S**)



41

41



42

The Introduction of Autonomous Systems

- ❑ In DV and LS algorithms, we view the network simply as a collection of interconnected routers.
- ❑ A group of routers are under the same administrative control
- ❑ All routers executed the same routing algorithm to compute routing paths through the entire network
- ❑ The routers in an ISP, and the links that interconnect them, constitute a single AS

43



Routing within a Stub AS

- ⊕ There is only one exit point, so routers within the AS can use *default routing*.
 - ❖ Each router knows all Network IDs within AS.
 - ❖ Packets destined to another AS are sent to the default router.
 - ❖ Default router is the border gateway to the next AS.



44



Challenges of Autonomous Systems

- Scale
 - Hundreds of millions of routers, impossible to store all the routing information for each of other routers
 - The overhead required to broadcast connectivity and link cost updates among all of the routers would be huge
 - A DV algorithm that iterated among such a large number of routers would surely never converge
- Administrative autonomy
 - The Internet is a network of ISPs, with each ISP consisting of its own network of routers
 - An ISP generally desires to operate its network as it pleases or to hide aspects of its network's internal organization from the outside



45



Inter-AS Routing

Why do we need BGP: various reasons

- Intra-AS routing only considers how to send messages from the source node to the destination node efficiently, without taking other complex factors into account (e.g. politics)
 - Different ASs are affected by many factors (politics, economy, and security) . They forwards packets with selection. For example:
 - An ISP only provides service to its own customers
 - Commercial traffic can not be transmitted over the education network
- There is a demand for inter-AS routing to achieve the routing among different ASs based on different routing policy.

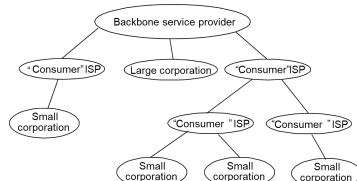
46



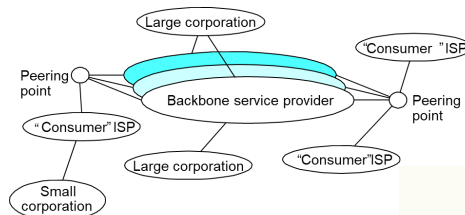
Inter-AS Routing

Why do we need BGP: Internet structure

- Original thought: network-centered tree topology



- Development: “networks of networks”



47



Inter-AS Routing

⊕ BGP: Border Gateway Protocol

- BGP allows each subnet to advertise its existence to the rest of the Internet
- It is the protocol that glues the thousands of ISPs in the Internet together



48

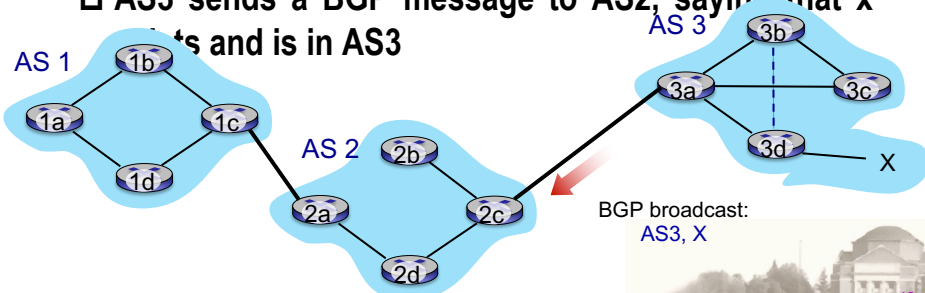
48



Inter-AS Routing

⊕ Advertising BGP route information

- BGP connection
 - Pairs of routers exchange routing information over semi-permanent TCP connections
- AS3 sends a BGP message to AS2, saying that x is in AS3



BGP broadcast:

AS3, X



49

49



Inter-AS Routing

⊕ Path attributes and BGP route

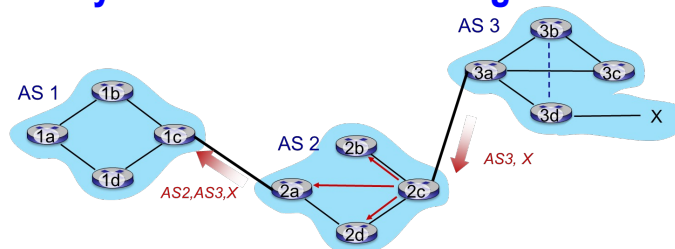
- BGP route: prefix+attributes
 - Prefix: indicate the destination
 - Two important attributes:
 - AS-PATH: contains the list of ASs through which the advertisement has passed
 - NEXT-HOP: the IP address of the router interface that begins the AS-PATH
- Policy-based routing
 - The routers that receives the BGP message determines the route according to the routing policy of their ASs
 - AS routing policy also determines whether advertises the reachability information to the neighbouring ASs

50



Inter-AS Routing

⊕ Reachability information advertising



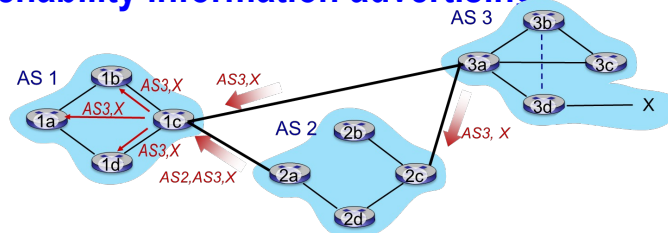
- Gateway router 3a sends an eBGP message “AS3 X” to gateway router 2c
- Based on the routing policy of AS 2, gateway router 2c then sends the iBGP message “AS3 X” to all of the other routers in AS2
- Based on the routing policy of AS 2, gateway router 2a then sends the eBGP message “AS2 AS3 X” to gateway router 1c

51



Inter-AS Routing

⊕ Reachability information advertising



□ A gateway router may know many different paths to a given destination

- Gateway router 1c is aware of the path “AS2 AS3 X” via router 2a
- Gateway router 1c is aware of the path “AS3 X” via router 3a
- Based on the routing policy, router 1c chooses the path “AS3 X” and uses iBGP to send the message “AS3 X” to all the routers in AS1

52



Inter-AS Routing

⊕ BGP message

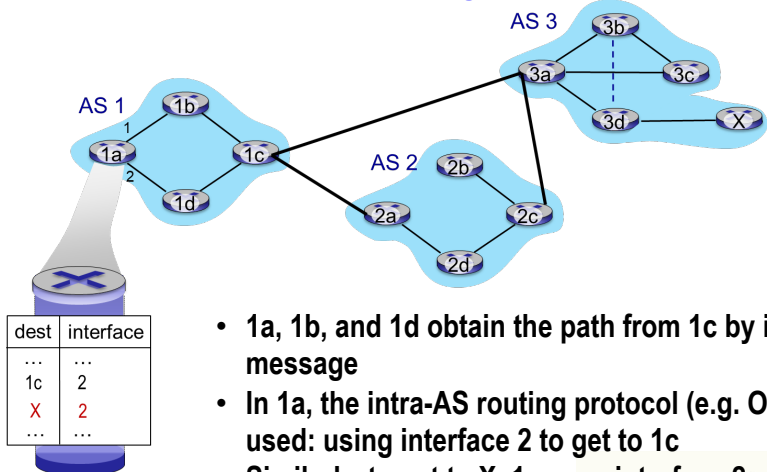
- BGP messages are sent between the BGP routers over TCP connections
- BGP message:
 - OPEN: establish BGP connections.
 - UPDATE: transfer routing information between BGP routers
 - KEEPALIVE: maintain BGP connections
 - NOTIFICATION: notify BGP routers of errors in a BGP process

53



Inter-AS Routing

⊕ BGP, OSPF, and forwarding table



dest	interface
...	...
1c	2
X	2
...	...

- 1a, 1b, and 1d obtain the path from 1c by iBGP message
- In 1a, the intra-AS routing protocol (e.g. OSPF) is used: using interface 2 to get to 1c
- Similarly, to get to X, 1a uses interface 2

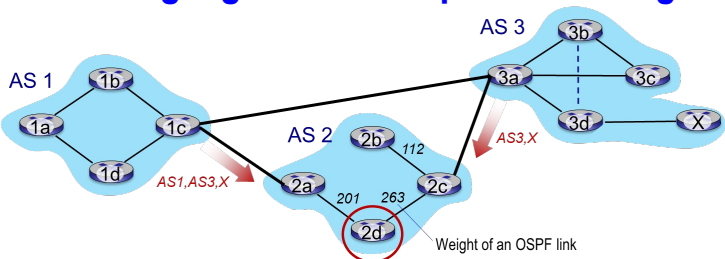

54

54



Inter-AS Routing

⊕ BGP routing algorithm: Hot potato routing



- ❑ 2d finds that it can get to X through 2a or 2c
- ❑ Hot potato routing: get packets out of its AS as quickly as possible, i.e., select the least-cost path to send the packet to another AS
- ❑ For example, 2d selects 2a though this path passes through more ASs to get to X

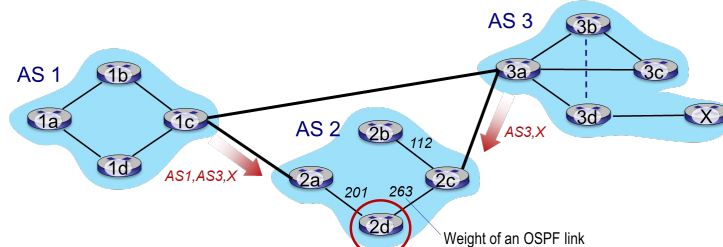

55

55



Inter-AS Routing

BGP routing algorithm: Hot potato routing



- ❑ 2d finds that it can get to X through 2a or 2c
- ❑ Hot potato routing: get packets out of its AS as quickly as possible, i.e., select the least-cost path to send the packet to another AS
- ❑ For example, 2d selects 2a though this path passes through more ASs to get to X

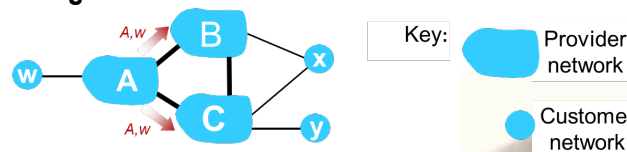
56



Inter-AS Routing

BGP routing policy

- ❑ Commercial ISPs believe that any traffic flowing across an ISP's backbone network must have either a source or a destination in a network that is a customer of that ISP
 - A advertises the path "A w" to B and C
 - B does not advertise "B A w" to C
 - Since C is not a customer of B, B can not have any benefits by advertising "B A w" to C
 - C does not learn the path "C B A w"
 - Packets from C will get to w by the path "C A w" without passing B



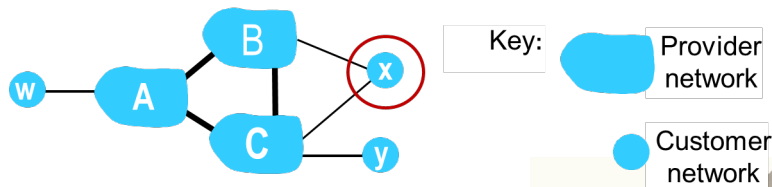
57



Inter-AS Routing

BGP routing policy

- A, B, and C are provider networks
- w, x, y are customer networks
- X is a multi-homed: it is connected to the rest of the network via two different providers
- Selective route advertisement policy: x does not want to advertise the path to any other destinations except itself
 - Thus, x will not advertise the path, say xC, to B



58



Inter-AS Routing

BGP route selection

- If there are two or more routes to the same destination, then BGP sequentially invokes the following elimination rules until one route remains:
- Local preference value assignment: policy decision (The routes with the highest local preference values are selected)
 - Shortest AS-PATH
 - Hot potato routing: the route with the closest NEXT-HOP is selected
 - Other BGP identifiers

59



Comparison between Intra and Inter-AS routing

⊕ Policy

- ❑ Intra-AS: single administrator, simple routing problem
- ❑ Inter-AS: administrators hope to control the routing of the traffic and the users they want to service

⊕ Scale

- ❑ Hierarchical routing reduces the size of forwarding table, which also reduces the traffic for update

⊕ Performance

- ❑ Intra-AS: focus on the performance
- ❑ Inter-AS: policy matters more than the performance

60



《通信与网络》
Communications & Networks

Thanks !

61